B. Two Divisors

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: constructive algorithms, math, number theory |
https://codeforces.com/problemset/problem/1916/B

A certain number $1 \le x \le 10^9$ is chosen. You are given two integers a and b, which are the two largest divisors of the number x. At the same time, the condition $1 \le a$ lt b lt x is satisfied.

For the given numbers a, b, you need to find the value of x.

dagger The number y is a divisor of the number x if there is an integer k such that $x = y \cdot k$.

Input

Each test consists of several test cases. The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. Then follows the description of the test cases.

The only line of each test cases contains two integers a, b ($1 \le a$ lt $b \le 10^9$).

It is guaranteed that a, b are the two largest divisors for some number $1 \le x \le 10^9$.

Output

For each test case, output the number x, such that a and b are the two largest divisors of the number x.

If there are several answers, print any of them.

Example

Input

```
8
2 3
1 2
3 11
1 5
5 10
4 6
3 9
250000000 500000000
```

Output

```
6
4
33
25
20
12
27
1000000000
```

Note

For the first test case, all divisors less than 6 are equal to [1, 2, 3], among them the two largest will be 2 and 3.

For the third test case, all divisors less than 33 are equal to [1, 3, 11], among them the two largest will be 3 and 11.

For the fifth test case, all divisors less than 20 are equal to [1, 2, 4, 5, 10], among them the two largest will be 5 and 10.

For the sixth test case, all divisors less than 12 are equal to [1, 2, 3, 4, 6], among them the two largest will be 4 and 6.